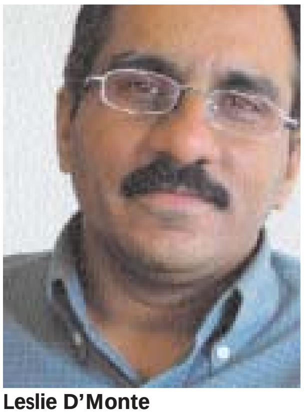
Computing sorely needs lateral thinking



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Physics and biology promise to provide a volcanic thrust to computing. Myriad concepts are floating around. Take for instance, Quantum computing, which uses traits of particles such as atoms and electrons. Theoretically, they are hundreds times faster than today's computers at solving very large problems. Where the digital computer uses binary digits (bits), the quantum computer uses quantum bits (qubits). Such a computer could resemble the coffee cup on your table.

Then there's DNA computing, which draws its power from the memory capacity and parallel processing. While a DNA computer takes much longer than a normal computer to perform each individual calculation, it performs an enormous number of operations at a time—massively parallel. DNA computers also require less energy and space than normal computers. A thousand litres of water could contain DNA with more memory than all the computers ever made, and a kilogram of DNA would have even more computing power.

Now bacteria are very cheap to manufacture. Scientists can grow trillions of bacteria in a laboratory for a few dollars as against the couple of billion dollars required to build a semiconductor fabrication plant. Little wonder that scientists have thought of bacterial computing too.

These computers may not do the same tasks as traditional computers such as surfing the Web. In all likelihood, they will be used in the study of logic, encryption, genetic programming and algorithms, and a plethora of other things that are currently beyond our imagination.

Even the network is benefiting. Researchers from Humboldt University in Germany have found that autonomous pieces of computer code can work like ants and bacteria. They have devised a way for electronic agents to assemble a network without relying on a central plan. The cue has been taken from insects and other life-forms whose communications lack central planning, but who manage to form networks when individuals secrete and respond to chemical trails.

It's not always physics, or biology, that is needed to give computing a tectonic shift. And, then, this is not what bothers end-users. These developments may merely be relegated to the area of intellectual gymnastics for the *junta*. What is sorely needed is a dose of lateral thinking. Chip giants have realised that end users need something more than mere power—that is, ease of use and fewer crashes to begin with. So they took a break from pumping adrenaline into their processors, and started thinking from the consumer point of view—about you and me.

So now you have the concept of autonomic computing, whereby PCs can protect, diagnose and heal themselves. Companies such as IBM are already making laptops that can be diagnosed and repaired with a simple press of a key. Network administrators are benefitting from this development, and the end-user is heaving a sigh of relief.

You also have the recent development called Centrino. It's after a major impassé that the chip giants are looking at mobile computing with the respect and attention it deserves. All this while, the chip makers such as Intel and AMD would simply take a desktop processor, reduce its frequency and voltage, and conveniently pass it off as a mobile processor. Little, or no, consideration was being paid to the battery life (a vital point) or weight. With no option, the user would have to lap it up. The laptop users among us should surely be glad to get some weight off our laps, and have a battery that can last 4 to 5 hours. Centrino, with its new architecture, has changed all that. Transmeta, with its Astro mobile processor, too has thought on similar lines.

Centrino mobile technology has also brought about an increased awareness of the advantages of being connected sans wires. Thanks to Centrino, we can get around the clutter of cables and think of WiFi (and later of 802.11a/g, and even ultra wideband) being a part of our motherboards. Just hit a WiFi hotspot (as in the U.S) and presto! You're connected to the world.

And as computing becomes as easy as switching on the lights or fan, we can revert to thinking about how we can use them. As for me, I just wish I could use my computer to post a gigantic mid-air message which reads: "War never decided who was right, only who was left".